Diffuse large B-cell lymphoma

DLBCL NOS · HGBCL with MYC + BCL2 rearrangements · PMBL · transformed · HIV-associated. **~60–70% are cured with first-line R-CHOP-based therapy** — key factors: double-hit status, IPI, CNS risk, fitness for anthracycline, and at relapse, timing (<12 vs ≥12 months) and CAR-T eligibility

Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin (Chou L, Schroers-Martin J, Gupta NK) and HIV-associated lymphoma · NCCN · primary trial reports and FDA actions verified against PubMed, September 2026.

PATHOLOGY REPORT

Call	Why it matters
Cell of origin (Hans IHC)	**GCB** = CD10+, or CD10– BCL6+ MUM1–. Everything else is **non-GCB (ABC-like)** — worse on R-CHOP, and the group where Pola-R-CHP gained most. GEP is more accurate than IHC
MYC / BCL2 / BCL6 FISH	MYC-rearranged in ~**10%** of aggressive B-NHL, half with BCL2. **MYC + BCL2 rearranged = HGBCL / DLBCL with MYC and BCL2 rearrangements** (formerly "double hit"). WHO-HAEM5 folds MYC + BCL6 back into DLBCL NOS / HGBL NOS; ICC keeps it provisional
Double expressor (IHC)	MYC ≥40% + BCL2 ≥50% protein **without** rearrangement — ~20–25%, intermediate prognosis, **treated as DLBCL NOS**. Not the same as double hit
CD30, EBER, CD5	CD30 → BV option at relapse · EBV+ DLBCL (any age, worse) · CD5+ enriched for CNS relapse

WORKUP

- **Excisional or core biopsy — never FNA**; hematopathology review
- **PET/CT** (staging + Deauville response); marrow biopsy only if it changes management — PET detects most marrow disease
- Labs: CBC, CMP, **LDH**, uric acid; **HBsAg + anti-HBc**, HCV, **HIV**
- **Echo or MUGA before anthracycline** · fertility counselling · pregnancy test
- **LP (cytology + flow) if CNS-IPI high** or high-risk site; MRI brain if symptomatic

IPI — APLES: AGE >60 · PS ≥2 · LDH ↑ · ≥2 EXTRANODAL · STAGE III–IV

Score	5-yr OS pre-R	3-yr OS R-CHOP
Low (0–1)	73%	**91%**
Low-intermediate (2)	51%	81%
High-intermediate (3)	43%	65%
High (4–5)	26%	**59%**

Shipp NEJM 1993 · Ziepert JCO 2010. **CNS-IPI** = IPI + kidney/adrenal (0–6): 2-yr CNS relapse **0–1: 0.6%** · 2–3: 3.4% · **4–6: 10.2%** (Schmitz JCO 2016).

TREATMENT PATHWAY

Limited stage (I–II, non-bulky)

R-CHOP ×4 is the backbone
FLYER: age ≤60, aaIPI 0, <7.5 cm → 4 R-CHOP + 2 R
S1001: R-CHOP ×3 → iPET; neg → 1 more (no RT)
iPET-positive (~11%) → ISRT
bulky or IPI-risk → treat as advanced

Advanced (III–IV, bulky, or IPI ≥2)

R-CHOP ×6 or Pola-R-CHP ×6
Pola-R-CHP for IPI ≥2 (POLARIX): PFS gain,
still no significant OS gain at 5 yr
HGBCL MYC+BCL2: DA-EPOCH-R (see p2)
PMBL: DA-EPOCH-R, or R-CHOP ± RT (see p2)

End-of-treatment PET/CT
Deauville 1–3 = CMR → observe
no maintenance, no consolidative ASCT in CR
Deauville 4–5 → **biopsy** before calling failure

Unfit for full R-CHOP

Age >80 or frail
R-mini-CHOP (2-yr OS 59%) — single-arm
do a geriatric assessment, prephase prednisone

LVEF low / prior anthracycline
R-CHOP + dexrazoxane + serial TTE (borderline)
R-GCVP · R-CEOP (inferior to R-CHOP) · R-CEPP
R-CDOP (liposomal doxorubicin)

Relapsed / refractory

Early (<12 mo) or primary refractory
CAR-T: axi-cel or liso-cel (ZUMA-7, TRANSFORM)
bridge with R-chemo, pola, or RT

Late (≥12 mo), transplant-fit
salvage R-ICE / R-DHAP / R-GDP → ASCT if CMR/PR
CAR-T is also reasonable — data are from <12 mo

Transplant-ineligible or 3L+
liso-cel (PILOT) · bispecific · pola-BR · tafa-len
BV-R² · loncastuximab · **trial** — see page 2

LIMITED STAGE — EVIDENCE

- **FLYER**: 4 R-CHOP + 2 rituximab non-inferior to 6 R-CHOP — 3-yr PFS **96% vs 93%**, a third less toxicity. **Only for age 18–60, aaIPI 0, bulk <7.5 cm**
- **SWOG S1001** (single-arm phase 2): R-CHOP ×3 → interim PET; **89% were PET-negative and got just 1 more cycle, no RT**. 5-yr PFS 87%, OS 89%; iPET-positive patients (RT) did as well
- **RT now:** bulk, PET-positive residual, or when chemo must be short

ADVANCED STAGE — EVIDENCE

- **R-CHOP is 20+ years old** (GELA LNH-98.5: 10-yr OS 43.5% vs 27.6% over CHOP); add-ons (ibrutinib, obinutuzumab, venetoclax) and DA-EPOCH-R have not beaten it
- **POLARIX (Pola-R-CHP)**: 2-yr PFS 76.7% vs 70.2%, HR 0.73. **5-yr update: PFS HR 0.77 (64.9% vs 59.1%), OS HR 0.85 — not significant**.
  Benefit concentrated in **ABC/non-GCB, IPI 3–5, age >60**. Needs G-CSF. Standard option for IPI ≥2
- **frontMIND** (tafa-len + R-CHOP, IPI 3–5): PFS HR 0.75, OS immature, more fatal AEs — **not FDA-approved**

CNS PROPHYLAXIS

- **Who is high risk:** CNS-IPI 4–6 · kidney/adrenal · **testicular** · breast, uterine · epidural / paranasal / skull base · HGBCL MYC+BCL2
- **HD-MTX does not clearly prevent CNS relapse.** 1,923 ultra-high-risk patients (Wilson JCO 2026): 3-yr CNS relapse **8.1% with vs 9.3% without HD-MTX**, adjusted HR 1.13 — no benefit in any subgroup. Alberta cohort: 11.2% vs 12.2%
- End-of-treatment timing is no worse than intercalated (5.8% vs 5.7%) and avoids R-CHOP delays. **Adding IT did not lower CNS relapse**
- **Practical:** image/LP high-risk patients at diagnosis; **testicular DLBCL still gets CNS-directed therapy + contralateral testis RT**; otherwise HD-MTX 3–3.5 g/m² ×2–4 is a shared-decision, not a default

STAGING — COTSWOLDS-MODIFIED ANN ARBOR

I	**One** lymph node region or lymphoid structure (spleen, thymus, Waldeyer ring)
II	**≥2** regions, **same side** of the diaphragm. Hilar nodes are lateralized — bilateral involvement is stage II
III	Regions on **both sides** of the diaphragm
IV	**Disseminated** — marrow, CNS, or **more than one** extranodal site
A / B	B = weight loss >10% over 6 mo, **or** unexplained fever >38°C in the past month, **or** recurrent drenching night sweats
S · X · E	S splenic · **X bulky: node ≥10 cm**, or mediastinal mass ≥⅓ the transverse thoracic diameter at T5–6 · E limited direct extranodal extension

A single extranodal site as the only disease is IE; more than one extranodal site is stage IV. Hilar (bronchopulmonary) nodes sit outside the mediastinum.

Diffuse large B-cell lymphoma

HGBCL WITH MYC + BCL2 REARRANGEMENTS

- **FISH every new DLBCL** — morphology and COO do not predict it. Often high LDH, marrow and CNS involvement
- **DA-EPOCH-R ×6** is the usual choice (single-arm, MYC-rearranged: 4-yr EFS 71%, OS 77%; Dunleavy 2018) — **no randomized proof it beats R-CHOP**; Pola-R-CHP is used by some (small POLARIX subset). **Evaluate the CNS** at diagnosis
- **No consolidative ASCT in first CR.** Relapse → CAR-T

PRIMARY MEDIASTINAL B-CELL LYMPHOMA (PMBL)

Who	Young (median ~30s), **women**, bulky anterior mediastinal mass ± SVC syndrome. Biology overlaps nodular-sclerosis cHL: **9p24.1 amplification** → **PD-L1**/**PD-L2**; CD30 often weakly +
DA-EPOCH-R ×6	NCI phase 2 (n=51): EFS **93%**, OS **97%** at 5 yr; only 2 of 51 needed RT. Avoids mediastinal RT (breast cancer, cardiac disease) in young women
R-CHOP ×6 ± RT	Accepted alternative. **IELSG37**: if end-of-chemo PET is **CMR (Deauville 1–3), omit RT** — 30-mo PFS 96.2% observation vs 98.5% RT, 5-yr OS 99% both. Deauville 5 does poorly (5-yr PFS 60%)
Relapsed	**Pembrolizumab** (KEYNOTE-170: ORR 45%, CR 13%; accelerated approval) · **nivolumab + BV** (CheckMate 436: ORR 73%, CR 37%; not FDA-approved for PMBL) · CAR-T (included in LBCL labels) · salvage + ASCT if chemosensitive

Mediastinal gray-zone lymphoma (PMBL/cHL features, often CD15+, CD20 variable): worse than either; DA-EPOCH-R, ISRT if localized.

HIV-ASSOCIATED AGGRESSIVE B-CELL LYMPHOMA

Principles	**Continue or start ART through chemotherapy** (check interactions — avoid boosted PIs/cobicistat with vincristine). PJP (TMP-SMX), **MAC prophylaxis if CD4 <100**, fluconazole, acyclovir; consider quinolone at nadir if CD4 <100
DLBCL	**R-CHOP or DA-EPOCH-R ×6** (with ART, outcomes approach HIV-negative). AMC-034: concurrent R-EPOCH CR **73%** vs sequential 55%. **Rituximab caveat: CD4 <50** — AMC-010 infectious deaths 14% vs 2% (60% of them at CD4 <50)
Plasmablastic	**CD20–**, CD138+, MUM1+, EBV+ in most, MYC often rearranged; oral cavity. **Rituximab useless** — DA-EPOCH (± bortezomib, small series); poor outcomes
PEL	**HHV8+** (most EBV+), **effusions without a mass**, CD20–, CD30+, CD138+. Modified EPOCH (NCI: median OS 22 mo); IL-6 high = worse. Exclude KSHV-MCD/KS
CNS prophylaxis	IT MTX for marrow, paranasal sinus, testicular or epidural involvement (handbook): 12 mg d1+5 of cycles 3–5, or weekly ×4 in cycle 1

RELAPSED / REFRACTORY DLBCL — AGENTS, EVIDENCE, STATUS (SEPT 2026)

Setting · agent	Evidence	Status / caveat
2L, early relapse axi-cel	**ZUMA-7** vs salvage + ASCT: EFS 8.3 vs 2.0 mo (HR 0.40); **4-yr OS 54.6% vs 46.0%, HR 0.73 — the only 2L CAR-T OS win**	Regular approval (Apr 2022). Gr ≥3 CRS 6%, neuro 21%
liso-cel	**TRANSFORM**: EFS NR vs 2.4 mo, CR 74% vs 43%; **OS HR 0.72, NS** (crossover-adjusted HR 0.42)	Approved 2L (Jun 2022), incl. transplant-ineligible (PILOT ORR 80%, median age 74). Gr 3 CRS 1%
tisa-cel	Negative in 2L (BELINDA)	**3L+ only**
2L, late relapse salvage + ASCT	PARMA (pre-R): ASCT 5-yr EFS 46% vs 12%. **CORAL**: R-ICE = R-DHAP (ORR ~63%); 3-yr EFS **20% if relapse <12 mo** vs 45% if later	Standard for chemosensitive late relapse; 2–3 cycles then BEAM-ASCT
Transplant-ineligible glofit-GemOx	**STARGLO**: OS 25.5 vs 12.9 mo (HR 0.62) vs R-GemOx; largely ex-US	**Not US-approved** (FDA CRL Jul 2025); NCCN option
epco-GemOx	EPCORE NHL-2 (single-arm): ORR 85%, CR 61%, median OS 21.6 mo	Not FDA-approved. **EPCORE DLBCL-4** (epco + len vs R-GemOx) PFS positive, topline Jun 2026
pola-BR	GO29365: CR 40% vs 18%; OS 12.4 vs 4.7 mo — **only 40 per arm**	Approved 2019. **Bendamustine impairs later T-cell collection** — avoid before CAR-T apheresis
tafa-len	L-MIND single-arm: ORR 57.5%, CR 41%, mPFS 11.6 mo (5-yr) — best after 1 prior line	**Still accelerated**; CD19 — not before CD19 CAR-T is a real concern
BV + R²	**ECHELON-3** vs placebo + R²: OS 13.8 vs 8.5 mo, HR 0.63 — benefit regardless of CD30	Approved Feb 2025, 3L+, **CAR-T and ASCT-ineligible**
3L+ CAR-T	ZUMA-1 (axi-cel) ORR 82%, CR 54% · TRANSCEND (liso-cel) ORR 73%, CR 53% · JULIET (tisa-cel) ORR 52%, CR 40%	All approved 3L+; high-risk, chemo-refractory populations
glofitamab	NP30179: CR 39% (35% post-CAR-T), **fixed 12 cycles**; median CR duration ~30 mo	**Accelerated**, ≥2 lines. Obinutuzumab pretreatment
epcoritamab	EPCORE NHL-1: ORR 63%, CR 39%, SC, until progression	**Accelerated**, ≥2 lines; **confirmatory EPCORE DLBCL-1 missed OS** (HR 0.96)
loncastuximab	CD19-PBD ADC. LOTIS-2: ORR 48%, CR 24%. **LOTIS-5** (lonca-R vs R-GemOx): PFS 6.1 vs 4.7 mo, HR 0.73; OS HR 0.96	Accelerated; confirmatory positive on PFS (Jun 2026). Edema/effusions, photosensitivity, GGT
selinexor	SADAL: ORR 28%, single-arm	**DLBCL indication withdrawn Apr 2026**

Sequencing: CAR-T first when feasible — bispecifics work after CAR-T (glofit CR 35%), but CAR-T after CD19-directed therapy (tafasitamab, lonca) is less certain; confirm CD19 by IHC. Avoid bendamustine before apheresis.

CHOOSING THERAPY AT FIRST RELAPSE

- **Biopsy the relapse** — confirm LBCL, check CD19 (before CD19-directed agents) and CD20 (lost in some after anti-CD20 therapy)
- **Timing is the strongest predictor.** In CORAL, relapse <12 months after rituximab-containing therapy gave 3-yr EFS of only 20% with salvage + ASCT — this is why CAR-T replaced ASCT for early relapse
- **CAR-T eligibility is about organ function and performance status more than age** — PILOT treated a median age of 74. Refer at relapse, not after failed salvage
- **Bridging:** steroids, RT to bulky sites, pola-R ± bendamustine-free chemo; avoid bendamustine before apheresis
- **After CAR-T failure:** bispecific (glofitamab fixed-duration, epcoritamab continuous), BV-R², loncastuximab, allo-HCT in fit chemosensitive patients, trial

TRANSFORMED INDOLENT LYMPHOMA

- **Biopsy the fastest-growing / highest-SUV site** (SUV discordance, rising LDH, B symptoms)
- **Anthracycline-naive → treat as de novo DLBCL** (R-CHOP or Pola-R-CHP). Anthracycline-exposed → manage as R/R DLBCL
- **CAR-T and bispecific labels include DLBCL arising from follicular/indolent lymphoma**. Richter (from CLL) is different — see the CLL card

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FRONTLINE DOSING

Regimen	Dosing (mg/m² unless stated)
R-CHOP q21d ×6	Rituximab 375 · cyclophosphamide 750 · doxorubicin 50 · vincristine 1.4 (cap 2 mg) — all d1; prednisone **100 mg** d1–5 (handbook: 40 mg/m²). G-CSF if age ≥60
Pola-R-CHP q21d ×6 (+2 R)	**Polatuzumab 1.8 mg/kg** replaces vincristine; R 375 · C 750 · H 50 d1; prednisone 100 mg d1–5. **G-CSF required**
DA-EPOCH-R q21d ×6	R 375 d1; **etoposide 50 + doxorubicin 10 + vincristine 0.4 /day CIV d1–4** (96 h); cyclophosphamide 750 d5; prednisone 60 **BID** d1–5 (NCI protocol; handbook lists 60/day); G-CSF from d6. CBC twice weekly: ANC nadir ≥500 → **+20%** (E, H, C); <500 on 3 checks or plt <25k → –20%
R-mini-CHOP q21d ×6	R 375 · C **400** · H **25** · V **1 mg flat** d1; prednisone 40 d1–5
R-GCVP q21d	R 375 d1; gemcitabine 1000 d1+8; C 750 + V 1–1.4 d1; prednisolone 100 mg d1–5 — anthracycline-free
HD-MTX (if used)	3–3.5 g/m² ×2–4 after R-CHOP; alkalinize urine pH >7, leucovorin rescue, levels to <0.1 µM; no third-spacing, no PPI/NSAID/TMP-SMX

DEAUVILLE 5-POINT SCORE

1 · 2	No uptake · uptake **≤ mediastinum**
3	Uptake **> mediastinum but ≤ liver** — usually adequate response
4 · 5	Moderately **> liver** · markedly > liver or new lesions — **treatment failure**

Response is reported this way in practice and in every trial cited. (The handbook itself specifies only PET/CT at completion, plus marrow biopsy if positive at diagnosis.)

CRS AND ICANS (ASTCT 2019)

Grade	CRS (fever ≥38 + …)	ICANS (ICE score)
1	Fever alone	ICE 7–9, awakens spontaneously
2	Hypotension without pressors and/or O_2 <6 L NC	ICE 3–6, wakes to voice
3	One pressor and/or high-flow / face-mask O_2	ICE 0–2, wakes to touch; focal edema, seizure resolving
4	Multiple pressors and/or positive pressure	Unarousable, status, diffuse edema

CRS: tocilizumab 8 mg/kg (max 800 mg) from grade 2 (earlier for CAR-T), add dexamethasone 10 mg q6h if refractory or grade 3+. **ICANS:** dexamethasone 10 mg q6h from grade 2; **tocilizumab does not treat isolated ICANS** (may worsen it); levetiracetam, neuro checks. Bispecific CRS clusters in cycle 1 after step-up doses.

SALVAGE AND R/R DOSING

Regimen	Dosing
R-ICE q14–21d	R 375 d1; **ifosfamide 5000 + mesna 5000 CIV 24 h d2**; carboplatin AUC 5 (max 800 mg) d2; etoposide 100 d1–3
R-DHAP q21d	R 375 d1; cisplatin 100 CIV d1; **cytarabine 2000 q12h ×2 d2** (1000 if >60; steroid eye drops); dexamethasone 40 mg d1–4. (R-DHAX: oxaliplatin 100)
R-GDP q21d	R 375 d1; gemcitabine 1000 d1+8; dexamethasone 40 mg d1–4; cisplatin 75 d1 — outpatient, non-inferior to R-DHAP (LY.12)
Pola-BR q21d ×6	Polatuzumab 1.8 mg/kg + R 375 d1; bendamustine 90 d1+2
Tafa-len q28d	Tafasitamab 12 mg/kg C1 d1, 4, 8, 15, 22 → C2–3 weekly → C4+ d1+15; lenalidomide 25 mg d1–21 ×12 cycles, then tafasitamab alone
Loncastuximab q21d	**0.15 mg/kg ×2 cycles, then 0.075 mg/kg**; dexamethasone 4 mg BID d–1 to d1
BV-R² q21d	BV 1.2 mg/kg (max 120 mg) + rituximab q3wk + lenalidomide 20 mg daily
Glofitamab IV q21d ×12	**Obinutuzumab 1000 mg C1D1 → 2.5 mg C1D8** (inpatient 24 h) → 10 mg C1D15 → 30 mg C2–12. Dexamethasone 20 mg IV before C1–3 doses
Epcoritamab SC q28d	**0.16 mg C1D1 → 0.8 mg C1D8 → 48 mg C1D15, D22**; weekly C2–3, q2wk C4–9, q4wk C10+. Dexamethasone 15 mg (or prednisolone 100) on dosing day + 3 days after in C1. C1D15: **outpatient monitoring allowed since Apr 2026**
CAR-T (single infusion)	**Axi-cel** 2×10⁶/kg (max 2×10⁸) after flu 30 + cy 500 ×3 d · **Liso-cel** 90–110×10⁶ (2L) after flu 30 + cy 300 ×3 d · **Tisa-cel** 0.6–6×10⁸ after flu 25 + cy 250 ×3 d

TOXICITY AND MONITORING

- **Hepatitis B with every anti-CD20** (and bispecifics): HBsAg+ → entecavir/tenofovir; **HBsAg– anti-HBc+ → prophylaxis** (or monthly DNA); continue **≥12 months after the last dose**
- **Cardiac:** baseline echo; R-CHOP ×6 = doxorubicin 300 mg/m². Risk rises steeply past 400–450 mg/m² — dexrazoxane or anthracycline-free regimen when EF borderline
- **Tumor flare** with bispecifics (and lenalidomide): painful node swelling, can compress airway/organs — steroids, antihistamine, hold
- **Infection:** acyclovir + PJP prophylaxis with bispecifics, CAR-T, pola-BR, EPOCH; **IVIG if IgG <400** with infections; CMV monitoring on bispecifics
- **CAR-T:** REMS removed Jun 2025 — stay near center **2 weeks**, no driving 2 weeks. Prolonged cytopenias; **secondary T-cell malignancy** boxed warning
- **Neuropathy** from vincristine (cap 2 mg) and polatuzumab; **TLS** prophylaxis for bulky disease or high LDH

KEY TRIALS

GELA LNH-98.5	R-CHOP vs CHOP, age 60–80 — 10-yr OS 43.5% vs 27.6%
POLARIX	Pola-R-CHP: PFS HR 0.73 → 0.77 at 5 yr; **OS HR 0.85, NS**
FLYER / **S1001**	Limited stage: 4 cycles enough; PET-adapted, RT for iPET+ only
ZUMA-7	Axi-cel 2L: EFS HR 0.40, **OS HR 0.73**
TRANSFORM	Liso-cel 2L EFS HR 0.35 (OS NS)
STARGLO	Glofit-GemOx OS HR 0.62 — US CRL (applicability)
ECHELON-3	BV + R²: OS 13.8 vs 8.5 mo
IELSG37	PMBL in CMR: RT omission non-inferior
Wilson 2026	HD-MTX: no CNS-relapse reduction in 1,923 ultra-high-risk
R-mini-CHOP	Age >80, single-arm: 2-yr OS 59%
NCI DA-EPOCH-R	PMBL EFS 93% without RT · MYC-rearranged 4-yr EFS 71%
PARMA / **CORAL**	ASCT beats salvage alone / R-ICE = R-DHAP; early relapse fares poorly
ZUMA-1 / **JULIET** / **TRANSCEND**	3L+ CAR-T: CR 54% / 40% / 53%
PILOT	Liso-cel 2L, transplant-ineligible: ORR 80%
NP30179 / **EPCORE NHL-1**	Glofitamab CR 39% (fixed) / epcoritamab CR 39% (continuous)
GO29365 / **L-MIND**	Pola-BR OS 12.4 vs 4.7 mo / tafa-len ORR 58%, CR 41%
LOTIS-2 / **LOTIS-5**	Lonca ORR 48% / lonca-R vs R-GemOx PFS HR 0.73
KEYNOTE-170 / **CM436**	R/R PMBL: pembro ORR 45% / nivo-BV ORR 73%
AMC-010 / **AMC-034**	HIV: rituximab deaths at CD4 <50 / concurrent R-EPOCH CR 73%